Enhancing Self-Regulated Learning Using Generative AI: Development and Evaluation of *SRLMentor*

Sikai Wang, Xinyi Luo, and Khe Foon Hew

***Abstract*—Effective self-regulation is crucial for student success; however, many students struggle with it, especially during online activities, leading to disengagement. Existing methods to boost self-regulated learning (SRL) skills, such as writing self-reflective reports and using prompts in video lectures, lack timely personalized feedback and can be labor intensive. This study implemented the Self-Regulated Learning AI Mentor (*SRLMentor*), a system designed to support students' SRL skills, including goal-setting, planning, help seeking, and reflection. *SRLMentor* comprises three integrated modules to provide timely customized SRL feedback to each student. The first module features an SRL knowledge base that stores user chat records and relevant memory-driven adaptive SRL prompts, addressing a significant limitation of large language models—inability to store new experiences in long-term memory during a dialogue. The second module incorporates a retrieval augmented generation (RAG) component to reduce content hallucinations, ensuring that students receive accurate information. The third module provides in-context learning examples that instruct the AI-based chatbot system to produce relevant SRL responses. We evaluated *SRLMentor* in an eight-session course with 25 students. Our assessment focused on RAG's performance in terms of factual consistency, answer correctness, and semantic similarity; accuracy of *SRLMentor's* detection of students' goals and plans compared to human coders; quality of *SRLMentor's* feedback; and students' perceptions of the system's usefulness. The results revealed that RAG enhanced factual, correctness, and semantic accuracy of responses. In addition, *SRLMentor's* assessments of students' goals and plans closely matched those of human coders. The cluster analysis revealed that students who engaged more with *SRLMentor* exhibited greater improvement in SRL skills and course knowledge compared to those who engaged less frequently with the system.**

***Index Terms*—Chatbots, generative AI, large language models (LLMs), online learning, self-regulated learning (SRL).**

I. Introduction

AFTER the end of the COVID-19 pandemic, many college students are opting for online classes. Online learning is expected to grow at a rate of 20.5% annually, with revenues surpassing US$1 trillion by 2030 [1]. This learning mode is essential for blended and hybrid approaches. As blended and hybrid learning methods gain popularity in the postpandemic era [2], the adoption of online learning activities continues to increase. Although online learning affords students enhanced flexibility, studies have reported lower student engagement levels in online environments than in in-person ones [3], [4]. Students require strong self-regulated learning (SRL) skills to effectively engage in online environments [5]. SRL is particularly important for online activities [6], as students have greater control over when, how, and where they learn [7]. A lack of SRL can lead to unrelated online activities [8], which can negatively impact learning.

SRL is an iterative process that includes various phases and subprocesses [9]. The most cited SRL model is Zimmerman's [10] three-phase model, which is widely recognized and offers a clear structured approach to understanding SRL. It is easy to understand and apply in classroom settings [9]. This model consists of forethought, performance, and self-reflection stages that continuously cycle to enhance learning outcomes.

During the forethought phase, students engage in goal setting and planning. Specific achievable goals lead to a consistent performance [11], whereas unrealistic goals can result in disappointment for students with weak self-regulation skills [12]. Goal setting and planning are intertwined, as students initially define their learning objectives and then devise strategies to accomplish them [13]. In the performance phase, students engage in academic tasks and monitoring, which involves being aware of and adapting their actions [14]. Help seeking is an important process [14]. Self-regulated students are not passive learners; rather, they actively seek help when needed [15]. Help seeking requires learners to know what, when, and whom to ask, reflecting a type of ongoing monitoring. The self-reflection phase involves self-evaluation, in which students evaluate and reflect on their learning performance concerning their goals [14], [16]. This phase is critical, as it informs the forethought phase for the next cycle. Based on their performance evaluations, individuals may revise their goals, plans, and performance, creating a dynamic feedback loop aimed at achieving better outcomes. The iterative nature of this model ensures that each phase builds upon the previous one, facilitating continuous improvement in SRL.

Students' ability to self-regulate their learning can positively impact their academic performance [17]. Thus, training in SRL

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the University of Hong Kong Human Research Ethics Committee.

The authors are with the Faculty of Education, The University of Hong Kong, Hong Kong (e-mail: sikaiw@connect.hku.hk; xylar@connect.hku.hk; kfhew@hku.hk).

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is crucial for enhancing engagement in online environments and can be implemented in three ways: direct instruction, integration into online activities, and trace data analysis. Direct instruction typically involves explicit SRL training *before* the course begins [6]. Bellhäuser et al. [18] used three lessons on Moodle covering goal setting to summarize key points with videos, presentations, self-tests, exercises, and group discussions. Students maintained daily diaries, significantly improving their SRL knowledge and behavior. However, this approach requires extra time beyond regular coursework [6], which may lead to student burnout [19].

The second approach involves integrating SRL support into online activities using prompts or SRL-focused questions, such as "What are your goals when learning from this video?" Van Alten et al. [20], [21] incorporated SRL prompts into video lectures, requiring students to answer questions to continue watching. However, the effectiveness of prompts varies, possibly due to the lack of customization in the hints provided, which are generic and not tailored to students' SRL levels [5].

The third method uses trace data to suggest subsequent actions. Researchers have used trace data to analyze students' SRL processes [22], [23]. Trace data comprise digital logs that capture user interactions with computer systems, software, or online platforms. These data provide a rich source of information for understanding SRL behavior. By applying a theoretical SRL model, researchers can map student actions onto specific SRL processes [22]. Metrics such as the average number of quiz retakes and completion rate of optional self-assessments can be collected. These metrics reflect the SRL reflection process, as they indicate students' efforts to review and reinforce their understanding of material [23]. Researchers have recently enhanced SRL detection by incorporating eye-tracking data, which improve the granularity of measurements [22].

Trace data can be used to provide personalized support for students' SRL. Lim et al. [24] developed a rule-based AI system using the conditional statement IF [condition] THEN [act]. The conditions were modeled from the trace data to detect SRL processes. The rule-based system tracked students' SRL in real time and provided personalized support when needed. For instance, if a student checked the learning goals and instructions but did not review the essay rubric within the first 2 min of a learning session, the system would prompt the student with a personalized scaffold, such as "Check the essay rubric" [24, p. 8]. However, relying on trace data to suggest subsequent actions may be limiting, as the system can provide recommendations only after a user has taken an action. This approach could frustrate users who prefer to wish to seek advice right from the start, *before* they take any action.

This study used large language models (LLMs) to address these challenges. We employed the Self-Regulated Learning AI Mentor (*SRLMentor*) system developed to support students' SRL skills, such as goal setting, planning, help seeking, and reflection. *SRLMentor* can be seamlessly integrated into online student activities, thereby avoiding the burden of an additional workload for students who would otherwise need to attend separate SRL training sessions before the start of a course.

II. Related Work

The emergence of LLMs, particularly ChatGPT-4, has attracted the global community. Initial research primarily explored LLMs' ability to answer test questions, their potential to support teaching and learning, and the possible challenges of using them. Some scholars posit that LLMs can also support students' SRL by providing personalized feedback [25].

However, a significant gap remains in the literature, which is dominated by speculative analyses concerning LLM's potential. Hennessy et al. [26] observed that numerous authors produce opinion pieces discussing AI's potential in learning and computing. The existing research on LLM-supported self-regulation is not immune to this issue. Despite numerous studies on this subject, many comprise nonempirical opinion pieces. Although some empirical studies have explored the potential use of LLMs to support student SRL, their results should be interpreted with caution.

First, not all studies clearly identify the SRL subprocesses targeted by ChatGPT or apply a particular SRL theoretical model to guide their system design. This hinders other researchers from replicating these studies. For instance, Delima et al. [27] used ChatGPT during mathematics lessons; however, the specific SRL subprocesses supported by the LLM were not explained.

Second, many empirical studies do not directly measure the impact of LLMs on student SRL [28], [29], [30], [31], [32]. For instance, Chiu [28] focused on teachers' opinions of ChatGPT's potential to foster SRL. Kong and Yang [30] assessed participants' knowledge and perceptions of AI concepts. Yan et al. [32] measured student perceptions of interacting with ChatGPT. Although these studies are informative, they do not provide definitive conclusions regarding the effects of LLMs on SRL.

Third, the duration of LLM was short in some studies. For instance, the participants in Nguyen et al.'s [31] and Hu et al.'s [33] studies interacted with ChatGPT for only 30 and 100 min, respectively. The results of these short-term studies may lack reliability due to factors such as the novelty effect.

Fourth, although LLMs are prone to hallucinations [34], no study has addressed this issue. Hallucinations refer to nonsensical and undesirable texts generated by the model. Consequently, users may risk relying on incorrect information provided by LLMs.

We developed *SRLMentor* to address these limitations and provide a comprehensive tool for promoting SRL. *SRLMentor* targets specific SRL subprocesses, such as goal setting, planning, help seeking, and self-evaluation (see Fig. 1).

The system was designed based on Zimmerman's theoretical SRL model, thereby providing a clear framework for its design and usage. We directly measured the impact of *SRLMentor* on student SRL using a validated SRL questionnaire. *SRLMentor* was designed for long-term use rather than for brief interactions. The platform is structured to support ongoing learning journeys, reduce the novelty effect, and allow for a sustained impact on SRL. Moreover, *SRLMentor* incorporates retrieval augmented generation (RAG) to reduce the occurrence of hallucinations and ensure the quality of the generated content. Furthermore, we integrated in-context learning examples to instruct *SRLMentor* to

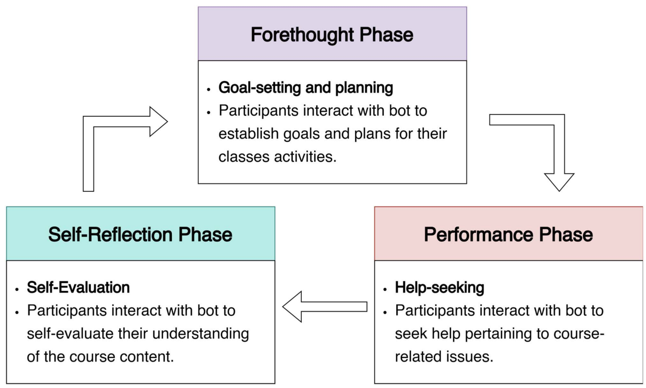

Fig. 1. Bots in various self-regulation phases.

produce relevant SRL responses. Therefore, *SRLMentor* offers a solution that addresses the limitations of previous studies and provides a comprehensive and effective tool for enhancing SRL.

A. Research Questions

This study addressed the following research questions (RQs).

RQ1: How well does RAG perform in terms of factual consistency, answer correctness, and semantic similarity?
RQ2: Can *SRLMentor* accurately recognize students' goals and plans, and how does its precision compare with human assessors?
RQ3: What is the quality of the overall feedback provided by *SRLMentor*?
RQ4: Does higher engagement with *SRLMentor* lead to greater improvements in students' SRL skills and learning performance?
RQ5: What is the students' perceived usefulness of the system?

III. System Architecture

SRLMentor is an advanced web platform that enhances SRL using a multiagent conversational system powered by LLMs. As shown in Fig. 2, its architecture includes three layers: 1) presentation; 2) data storage; and 3) business.

A. Presentation Layer

The presentation layer provides an engaging learning experience through a user-friendly interface. It features essential components for easy navigation and learner interactions. To achieve this, the Web front end utilizes advanced technologies, such as HTML5, CSS3, JavaScript, and Vue.js. This integration ensures a seamless and responsive user experience across various devices and platforms by effectively connecting the interface to the back end [35].

The user interface comprises several core elements. Chatbots (see Fig. S1 in Supplementary Material) foster cognitive processing, encourage critical thinking through follow-up questions, and enhance learning by promoting active engagement.

The SRL process monitor includes three components: Goal-Plan Checklist, Task Panel, and Notebook. The Goal-Plan Checklist (see Fig. S2 in Supplementary Material) helps students set clear objectives, strategize their learning approaches, and engage in proactive planning. The Task Panel (see Fig. S3 in Supplementary Material) allows students to upload reflective assignments provided by teachers, while the Notebook (see Fig. S4 in Supplementary Material) offers students space for annotating saved chat records. These tools empower learners to monitor their progress and reflect on their learning journeys, which are crucial for supporting SRL and improving learning outcomes.

B. Data Storage Layer

The data storage layer in *SRLMentor* stores and manages diverse data types, including user-generated content, application logs, and specialized SRL knowledge structures such as user goals, plans, notes, and SRL strategy usage. The architecture of this layer was designed to efficiently handle both structured and high-dimensional data. It uses a dual-database approach to optimize data retrieval and storage, ensuring that the platform operates smoothly and effectively.

SQLite, a lightweight relational database, is used to manage structured data due to its efficiency and suitability for environments that require compact and reliable solutions [36]. SQLite efficiently handles various types of structured data, such as user logs, that are captured by the user log recorder module.

The data storage layer captures qualitative data, such as chat history and user notes, and trace data, such as mouse clicks and chat duration. Recent studies have highlighted the importance of collecting online trace data, as they offer additional insights into students' SRL subprocesses [37]. By offering a comprehensive view of user SRL interactions and behaviors, the user log recorder facilitates the accumulation of authentic observations of students' actual interactions and learning activities, thereby providing a deeper understanding of users' SRL.

To manage high-dimensional vectorized data, such as document embeddings crucial for RAG, *SRLMentor* integrates Pinecone,[1] a vector database capable of efficiently handling large volumes of complex vector data. Pinecone's architecture is designed for fast and scalable access to vector data, which are essential for features that require quick retrieval and processing of embedded data representations [38]. The dual-database design, which incorporates both SQLite and Pinecone, offers a comprehensive data management solution that supports the complex and dynamic nature of online SRL environments. This setup ensures robust data integrity and access while effectively meeting the expanding data requirements of *SRLMentor.*

C. Business Layer

The business layer was created using a robust web back end developed using Python and the Django framework, known for its scalability and security features [39]. This back-end

[1] [Online]. Available: https://www.pinecone.io/

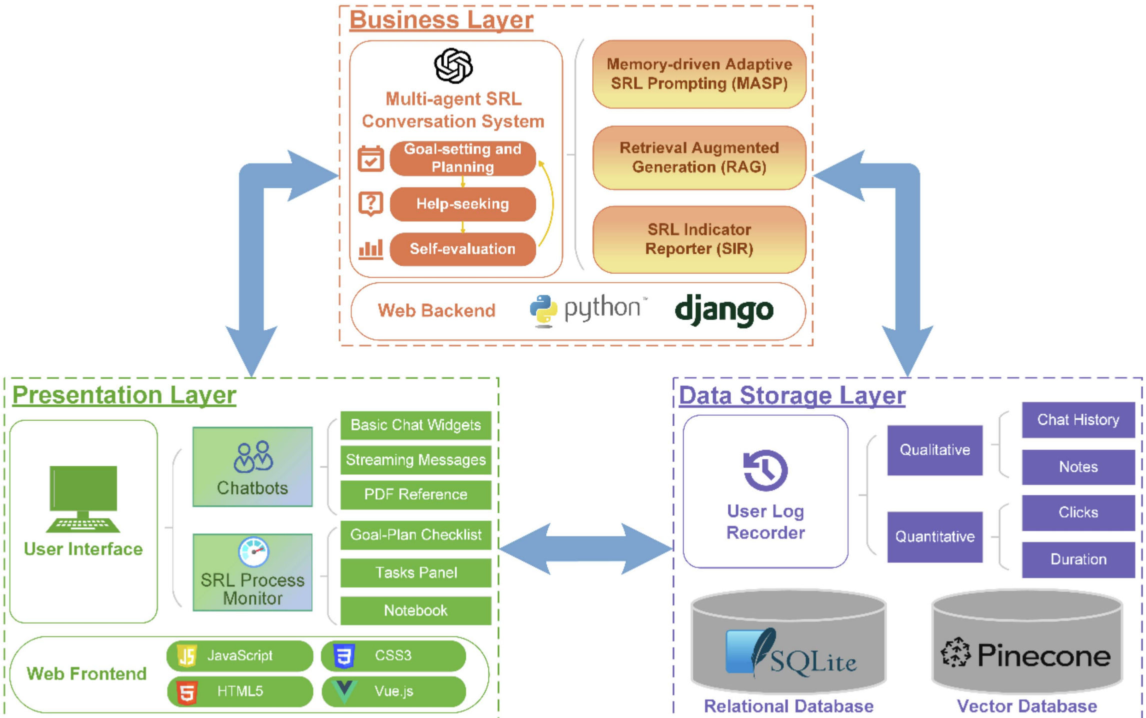

Fig. 2. High-level system architecture of *SRLMentor*.

architecture supports high-level functionality, ensuring seamless application programming interface (API) integration and reliable server-side logic handling.

It facilitates the processing of user inputs, session–state management, and execution of algorithms that support the SRL process. In addition, it supports a multiagent SRL conversation system that manages user interactions and aids in SRL activities, such as goal setting, planning, help seeking, and self-evaluation.

The key components of this layer are memory-driven adaptive SRL prompting (MASP) and RAG. These algorithms analyze user SRL characteristics from chat logs and databases to create personalized prompts, helping LLMs generate tailored prompts. SRL indicator reporter (SIR) continuously reports learners' engagement with SRL strategies.

1) Memory-Driven Adaptive SRL Prompting: The MASP method utilizes historical user data to tailor prompts according to learners' evolving needs and progress. This approach provides context-sensitive and adaptive SRL scaffolding through advanced prompt engineering. Fig. 3 illustrates the MASP process, which comprises three key steps.

First, *SRL context extraction* [see Fig. 3(a)] is used to operationalize SRL phases in a machine-readable format. We designed key SRL indicators to capture subprocesses from chat history and user log data. These indicators correspond to Zimmerman's three SRL phases.

1) Forethought phase: It captures learner-set goals and plans. These are initially input by the learner and periodically updated to reflect evolving learning objectives. For example, user chat statements like "I want to submit my project by the 25th" populate the goals field, while "I plan to allocate two days for revisions" updates the plans field.

2) Performance phase: It monitors learning tasks (e.g., project_timeline, project_monitoring, monitoring_strategies_comparison). For example, user chat statements such as "I need to finish the draft by next Friday" would update project_timeline, "I'm falling behind schedule" would populate project_monitoring, and "Using visual aids worked better than reading" would update monitoring_ strategies_ comparison. Submission time stamps from the Task Panel are also logged to calibrate the project_ timeline indicator. This ensures alignment between the planned progress (e.g., scheduled milestones) and actual progress (e.g., task completion dates).

3) Self-reflection phase: It focuses on learners' reflections on past learning experiences (e.g., draft_reflection, addressing_comments, project_satisfaction_rating, goal_completion, plan_completion). These indicators help identify areas for improvement. For example, user chat statements like "I'm satisfied with my work, and I would rate it a 4 out of 5" contributes to project_satisfaction_rating. In addition, user log actions such as marking goals and plans as completed in the Goal-Plan Checklist are recorded to populate the goal_completion and plan_completion indicators.

The system continuously monitors chat history and user logs and automatically extracts relevant SRL information using GPT-4's natural language processing (NLP) capabilities. Extracted data are then populated into corresponding SRL indicator fields in the JSON structure.

Second, *JSON structuring* [see Fig. 3(b)] is used. The extracted SRL indicators are stored in the JSON format, with each indicator tagged by its corresponding SRL subprocess. This structured representation ensures the following.

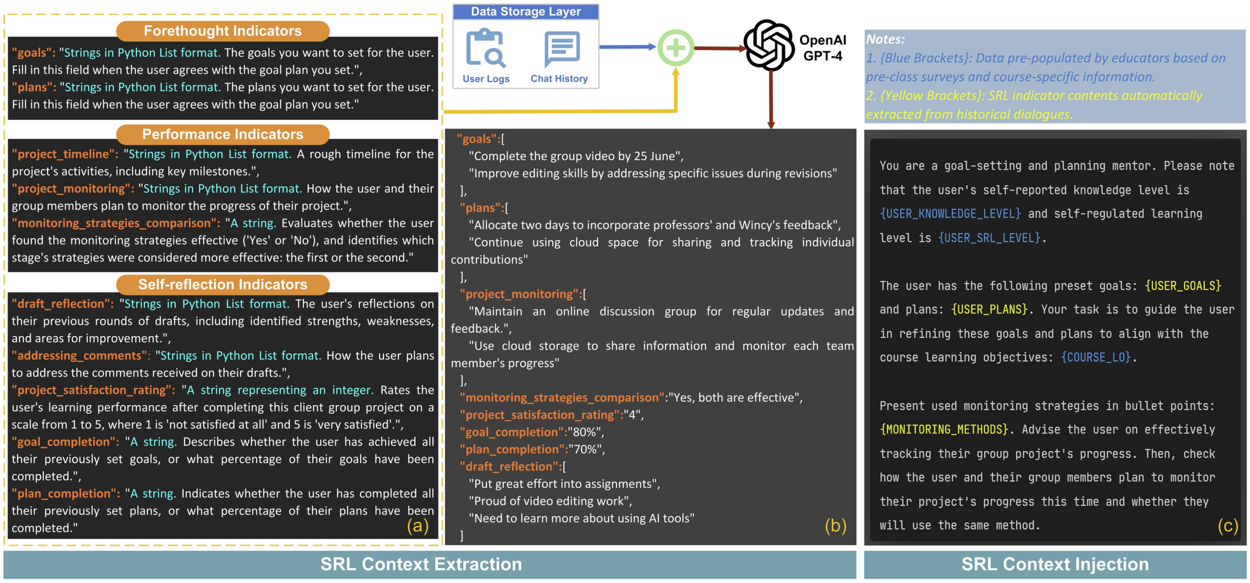

Fig. 3. (a)–(c) Memory-driven adaptive prompting.

1) *Machine readability:* The JSON format allows seamless integration with LLMs, enabling dynamic and automated prompt generation based on real-time learner data.
2) *Human auditability:* Educators and researchers can trace how the system interprets user input, identify the SRL indicators that drive adaptations, and evaluate the appropriateness of system responses.

Third, *SRL context injection* [see Fig. 3(c)] is used. MASP matches SRL indicators (by name) from the JSON structure and injects them into prompt templates authored by educators. This step ensures the following.

1) Prompts are informed by prior learner behaviors, and variables such as {USER_GOALS}, {USER_PLANS}, and {MONITORING_METHODS} are integrated into the prompt to reflect the learner's SRL progress.
2) The decision-making process is transparent, and by referencing specific SRL indicators, the MASP provides scrutable prompts for educators and researchers to evaluate.

The resulting prompt, enriched with SRL indicators, is then fed into the LLM-driven chatbot. This enables the chatbots to generate responses that are personalized and aligned with the SRL framework. For instance, the chatbot may detect mismatches between a learner's goals and progress, diagnose the underlying causes, and propose suitable adjustments.

MASP also incorporates educator-defined variables, such as {USER_SRL_LEVEL} (assessed using the Online Self-Regulated Learning Questionnaire, OSLQ [40]) and {USER_KNOWLEDGE_LEVEL} (assessed using knowledge tests), to guide the adjustment of prompt difficulty and content. For instance, a higher {USER_SRL_LEVEL} may encourage open-ended reflection, while a lower level may lead to more directive scaffolding. Similarly, users with advanced {USER_KNOWLEDGE_LEVEL} might receive domain-specific advice, whereas beginners are more likely to be offered step-by-step guidance. However, these variables do not strictly determine the LLM-driven chatbots' output; instead, they serve as guiding parameters to shape the interaction, allowing the chatbots to provide tailored support while retaining their flexibility and adaptability.

2) Retrieval Augmented Generation: Although traditional Generative AI chatbots can answer a wide range of questions, they often rely on static pretrained data, leading to inaccuracies. This can result in "hallucinations" when faced with unfamiliar or private data [41]. RAG addresses these limitations by accessing real-time domain-specific information from a knowledge base and providing accurate and contextually relevant responses [42]. This is particularly beneficial in SRL, in which help-seeking behavior is crucial. In *SRLMentor*, the RAG system enhances chatbot accuracy through a three-step process (see Fig. 4).

Step 1—Knowledge base construction: Educators upload course materials (PDFs), which are segmented and transformed into high-dimensional vectors using OpenAI's embedding model.[2] These vectors capture concepts within documents, thereby facilitating deep semantic analysis. These vectors are stored in the Pinecone database for real-time similarity searches.

Step 2—Retrieval: Upon receiving a student query, the system employs an LLM to break down the query into clearer subquestions. This process refines vague or disorganized inquiries, removes irrelevant details, and captures the user's true intent [43]. Despite the capabilities of newer LLMs to handle larger data volumes, overloading them can degrade performance and cause errors or hallucinations [44]. To mitigate these issues, our system incorporates a commercial reranking model, rerank-multilingual-v3.0[3] to assess document relevance. This model generates similarity scores based on how well documents align with the refined subquestions, ensuring that only the top-N most

[2][Online]. Available: https://platform.openai.com/docs/models/embeddings
[3][Online]. Available: https://docs.cohere.com/docs/rerank-2

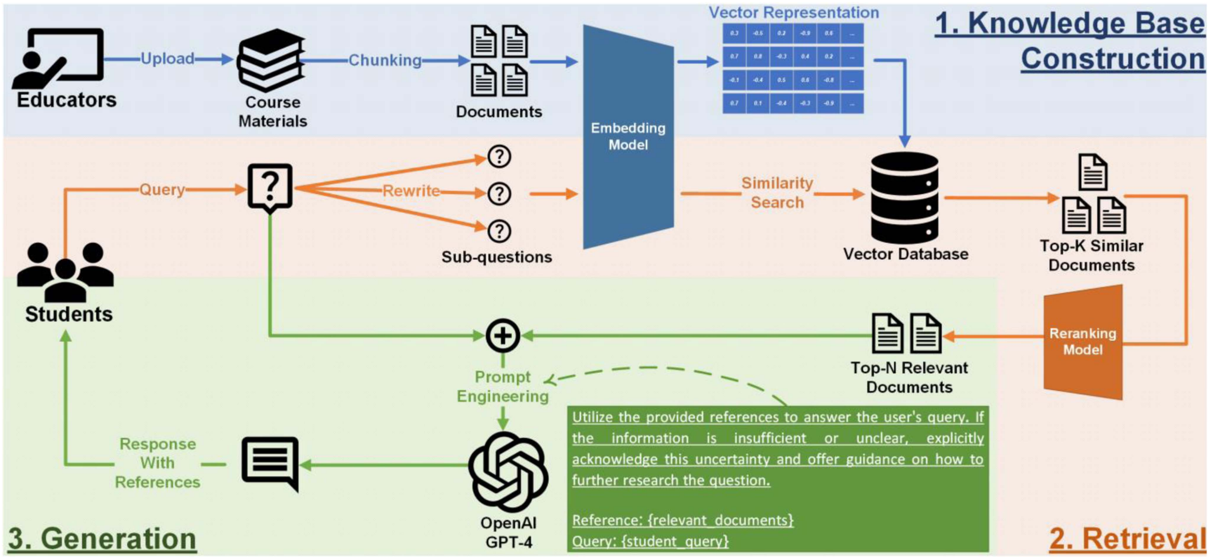

Fig. 4. Flowchart of the proposed RAG system in *SRLMentor*.

relevant documents are retained. This selection enhances the precision and recall of the retrieval operation, ensuring that chatbot outputs are accurate and relevant to the user's query.

Step 3—Generation: The system uses GPT-4 to synthesize a coherent response from N selected documents obtained in the previous step, including links to the original sources for verification. This enables *SRLMentor* to facilitate effective help-seeking behavior and precise query handling. As shown in Fig. S5a (Supplementary Material), the chatbot provides a concise summary of the project's scope and objectives from the course materials. This response includes an interactive feature (green reference link in the chat bubble). Clicking this link brings a user to a specific page in a preuploaded PDF (see Fig. S5b in Supplementary Material), ensuring precise answers and direct access to documents. If the content is insufficient or unclear, the system acknowledges these challenges and provides users with guidance for conducting further research or exploring additional resources related to the topic. Fig. S5c (Supplementary Material) shows the chatbot handling an off-topic query regarding Adobe Photoshop installation. It acknowledges its limitations, provides a general step-by-step guide, and directs the user to Adobe's website. This approach encourages users to seek information independently.

3) SRL Indicator Reporter: The SRL indicators extracted through MASP offer a detailed understanding of each learner's SRL attributes such as goals, plans, and strategies. SIR, a visualization tool, facilitates the reporting of these indicators with two key functionalities.

1) *Scheduled email dispatch:* SIR automatically sends personalized scheduled email reports containing SRL indicators. SIR-generated reports include specific questions from the chatbot and user responses (see Fig. S6 in Supplementary Material), allowing users to reflect on their interactions. This reflection enhances users' awareness of their own learning processes, strengths, and areas for improvement.
2) *Enhancement of Quizbot:* Quizbot is an interactive LLM-driven chatbot designed to assess students' understanding and mastery of course materials through quizzes (see Fig. S7 in Supplementary Material). Quizbot focused on *formative assessment*, providing immediate feedback to identify student knowledge gaps during the learning process.

After the quizzes are completed, SIR collaborates with Quizbot to present a table of responses, summarizing the quiz questions, the user's answers, the correct responses, and detailed explanations. Based on the quiz performance, SIR identifies key knowledge points and SRL strategies that require student attention (see Fig. S8 in Supplementary Material). Overall, by providing personalized feedback and actionable recommendations, SIR helps learners recognize their strengths and weaknesses, thereby enhancing their self-evaluation skills and strategic planning for future learning.

IV. Methodology

A. Overall Research Design

We conducted a field study to explore our RQs. Unlike laboratory studies, which occur in controlled environments over short durations, field studies occur in real-world settings, with participants engaging with a system for extended periods. The main advantage of field studies is their high ecological validity [45].

B. Participants and Procedure

The study included 25 postgraduate students (21 women and 4 men) between 20 and 40 years from an Asian public university. All participants were of Asian ethnicity. Written informed consent was obtained, and participants were assured that their participation was voluntary. Privacy and confidentiality were protected through anonymized data and secure storage. This

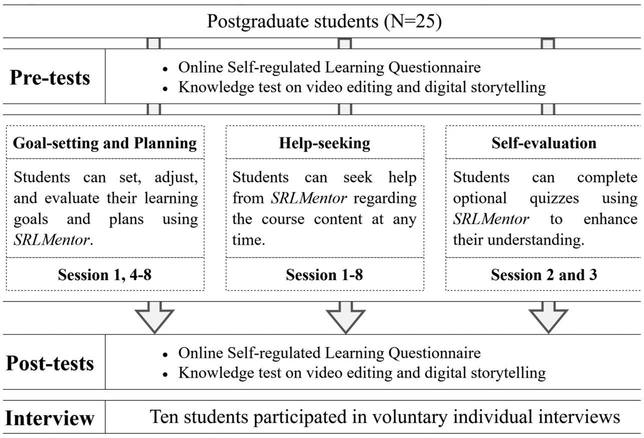

Fig. 5. Study procedure.

study was approved by the University Ethics Review Board. We conducted an eight-session course on video and digital storytelling in school and corporate settings combining online and in-person learning. Each face-to-face session lasted 3 h. Participants created short instructional videos as a course requirement. The study procedure is shown in Fig. 5.

In the first session, a researcher introduced the study and *SRLMentor*, assessed participants' initial knowledge of the course content, and administered a preactivity SRL survey. To familiarize students with *SRLMentor*, they were encouraged to set goals and plans for the entire course. To develop their goal-setting and planning skills, students further engaged in two rounds of goal setting and planning using *SRLMentor* during sessions 4–8. Specifically, the task involved editing a video to meet clients' needs. In the first round, students established specific goals and plans for the video editing task. Students revisited and adjusted their goals and plans during the second round based on their ongoing learning progress.

To enhance help-seeking skills, *SRLMentor* was equipped with RAG capabilities, allowing it to access all course-related materials. Students could interact with *SRLMentor* both during and outside class to seek assistance whenever needed. To foster self-evaluation, Quizbot was introduced, covering the learning content from sessions 2 and 3. Students were encouraged to complete the optional quizzes to assess their understanding of the course materials. In session 8, participants completed a postactivity SRL survey and postknowledge test. Following this, they were invited to participate in individual interviews. Table SI in Supplementary Material provides examples of participant interactions with *SRLMentor*.

C. Measures

1) Evaluation of the RAG Mechanism:: To evaluate the RAG mechanism, we used the Ragas framework [46], which employs automated algorithms to quantitatively assess RAG systems. We focused on three key metrics, each scored from 0 to 1, with higher values indicating better performance.

1) *Faithfulness:* This metric evaluates the factual consistency of generated content with the context on which it is based. The score is calculated as follows:

$$\text{Faithfulness} = \frac{C_{\text{inferred}}}{C_{\text{total}}} \tag{1}$$

where C_{inferred} represents the number of claims in the generated answer that can be inferred from a given context, and C_{total} denotes the total number of claims in the generated answer.

2) *Answer correctness:* This measure evaluates the accuracy of the generated answer by comparing it with the ground truth, considering both semantic and factual similarities. This metric is determined using the following formula:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

where true positive (TP) comprises facts or statements present in both the generated answer and ground truth, false positive (FP) comprises facts or statements present in the generated answer but not in the ground truth, and false negative (FN) comprises facts or statements present in the ground truth but omitted in the generated answer. We used the F1 score to assess answer correctness.

3) *Answer semantic similarity:* This metric evaluates how closely the meaning of the generated answer matches the ground truth regardless of the exact phrasing or structure. To calculate this, the ground truth and generated answer are converted into vectors using the same embedding model. The similarity between these vectors is measured using cosine similarity.

Based on these metrics, a comparative experiment was designed to evaluate the performance of the proposed RAG system. The study was divided into three groups.

1) *RAG+GPT-4:* This group used the RAG framework with the GPT-4 model, leveraging the advanced capabilities of the latest LLM (at time of writing).
2) *RAG+GPT-3.5:* This group was similar to the first group but used a slightly older GPT-3.5 model. This setup allowed us to directly compare the impact of the language model's inherent performance level on the retrieval-augmented output.
3) *Baseline GPT-4:* This control group received answers generated directly from GPT-4 using only the query, without an active retrieval step. Although pre-retrieved context from the dataset was used for metric calculations, this context was not utilized during answer generation. This nonaugmented baseline helped us examine the impact of retrieval augmentation.

We employed an algorithm from the Ragas Python APIs[4] to automatically construct a dataset from course materials in our knowledge base. Ragas leveraged an LLM to analyze

[4][Online]. Available: https://docs.ragas.io/en/stable/concepts/test_data_generation/rag

course documents to identify key concepts (e.g., topics and definitions) and their semantic relationships. These relationships were structured into a knowledge graph, mapping how concepts interconnect within the course content. Using the knowledge graph, Ragas systematically generated a set ($N = 200$) of triplets (Question, Context, and Ground Truth), forming our evaluation dataset. In these triplets: 1) Question represented simulated potential student inquiries; 2) Context represented relevant document segments from the knowledge base, serving as supporting evidence; and 3) Ground Truth represented the expected answers to each question which served as a benchmark for measuring response accuracy. To ensure validity, we manually verified that these triplets were accurately derived from the course content and reflected the correct knowledge.

2) Accuracy of Goal and Plan Detection: The accuracy of *SRLMentor* in recognizing student goals and plans and providing tailored feedback is crucial. To measure this, we compared the system's performance with that of human coders on a five-point Likert scale (1 = no accuracy; 5 = complete accuracy; see Table SII in Supplementary Material for examples).

3) Quality of SRLMentor Overall Feedback: Based on previous studies [47], [48], [49], [50], [51], we assessed the *SRLMentor* system's overall feedback quality using three criteria: coherence, relevance, and positive tone (see Table SIII in Supplementary Material for examples). The scores for these criteria were calculated as follows.

1) *Coherence:* This concept refers to the connection between sentences and ideas that helps readers better understand information [50, p.2]. Coherent feedback effectively links ideas, allowing readers to follow the thought process [49]. Coherence was rated on a three-point Likert scale (1 = incoherent; 2 = partially coherent; and 3 = coherent).
2) *Relevance:* Providing students with pertinent feedback on their work is crucial as it encourages them to improve both their current and future work [47]. Feedback was rated on a three-point Likert scale (1 = not related to objectives and strategies; 2 = contains irrelevant content; and 3 = aligns with goals and plans).
3) *Positive tone:* Feedback should encourage and convey a supportive tone [48], [51]. Positive feedback, such as praise, can motivate students to complete tasks [48]. Tone was rated on a three-point Likert scale (1 = negative; 2 = neutral; and 3 = positive).

4) Student SRL: We assessed students' SRL using the OSLQ [40], designed for online or blended learning settings. Self-report questionnaires can provide accurate insights into students' self-regulatory abilities [52]. The OSLQ consists of 24 items rated on a five-point Likert scale (1 = strongly disagree and 5 = strongly agree), with higher scores indicating better self-regulation. The OSLQ is widely accepted and used in various regions [40], [53], [53], [54], [55], [56], [57].

5) Learning Performance: To assess learning performance, pre- and postknowledge tests were administered. The pretest measured students' baseline familiarity with the course content, whereas the posttest assessed students' acquired knowledge at the end of the course. Both tests consisted of items covering key course concepts.

TABLE I
OVERVIEW OF LOG DATA FROM THE DATA STORAGE LAYER

Source	Log Variable	Description
Chat History	Turns_GoalPlanning	Number of utterance turns where the user engaged in goal setting and planning activities with chatbots such as defining goals and outlining steps to achieve them.
	Turns_HelpSeeking	Number of utterance turns where the user requests assistance or clarification from chatbots.
Notes	Notes_Taken	Number of times the user takes notes in the system.
Clicks	Checklist_Views	Number of times the user views the goal-plan checklist.
	Quizzes_Optional	Number of optional quizzes completed by the user.
Duration	Turns_Total	Total number of utterance turns during the user's interactions with all chatbots.

6) Student Engagement With the System: We categorized and aggregated raw system log data to capture student behavioral engagement in *SRLMentor* [58]. The relevant log data are summarized in Table I.

To determine whether differences in student engagement with *SRLMentor* were associated with variations in SRL skills and learning performance, we conducted a K-means cluster analysis on the log data. This data-driven approach enabled us to classify students into two groups characterized by low and high levels of behavioral engagement. Following this classification, we examined the differences in self-reported SRL scores and learning outcomes between the two groups.

7) Learning Perceptions: We interviewed ten participants to understand their perceptions of using *SRLMentor*. The questions covered their overall experience; how *SRLMentor* helped with setting, reflecting on, and monitoring personal goals and plans; suggestions for improvement; and effective methods of supporting learning progress monitoring.

V. RESULTS AND DISCUSSION

A. RQ1: How Well Does RAG Perform in Terms of Factual Consistency, Answer Correctness, and Semantic Similarity?

1) Faithfulness: As shown in Fig. 6, RAG+GPT-4 demonstrated the highest mean faithfulness score (0.78), followed by RAG+GPT-3.5 (0.71) and Baseline GPT-4 (0.17). This result highlighted the limitations of stand-alone language models in maintaining factual consistency without the aid of contextual data retrieval. The high faithfulness scores achieved by RAG+GPT-4 and RAG+GPT-3.5 suggested that the integration of a retrieval system significantly enhanced the factual alignment of the generated content with the provided context. This outcome may be attributed to the retrieval system's ability to provide the language model with relevant contextual data. By supplementing the model with accurate and context-specific information, the system likely reduced the chances of generating incorrect or inconsistent responses.

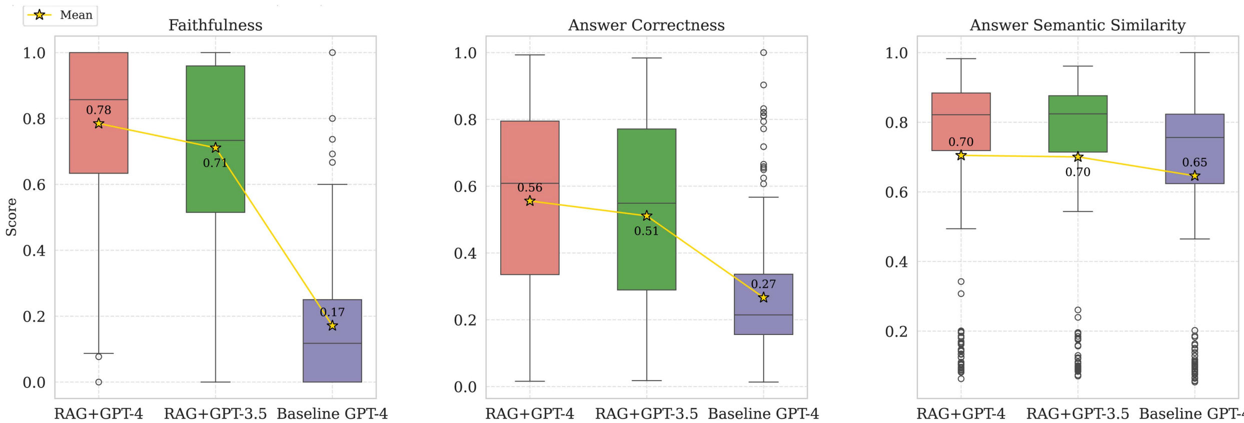

Fig. 6. Comparison across three groups.

By contrast, the Baseline GPT-4's low faithfulness score highlights the challenges that stand-alone models face in maintaining factual consistency without external data support. This underscores the importance of the retrieval component, which significantly bolsters the model's ability to produce contextually appropriate and factually accurate responses.

2) Answer Correctness: RAG+GPT-4 achieved the highest mean correctness score (0.56), followed by RAG+GPT-3.5 (0.51) and Baseline GPT-4 (0.27).

Despite using a powerful LLM, the absence of a retrieval framework significantly reduced the credibility of the generated content. Correctness scores indicated that integrating language models with retrieval systems significantly enhanced the accuracy of the answers provided. The consistent improvement across RAG+GPT-4 and RAG+GPT-3.5 suggested that the retrieval component ensured response accuracy regardless of the language model version. This improvement could be attributed to the retrieval system's ability to provide precise and relevant information, which the language model used to generate more accurate responses. The Baseline GPT-4's lower correctness score underscores the limitations of relying solely on the language model without the benefit of external data retrieval, leading to less credible and accurate content.

3) Answer Semantic Similarity: RAG+GPT-4 and RAG+GPT-3.5 achieved mean scores of 0.70 in semantic similarity, followed by Baseline GPT-4 (0.65). The similar semantic similarity scores for RAG+GPT-4 and RAG+GPT-3.5 suggested that the retrieval-augmented systems maintained a high level of semantic coherence, regardless of the underlying language model. The retrieval component effectively enhanced the semantic alignment of the generated responses with the ground truth. This slight improvement over the Baseline GPT-4's semantic similarity score can be attributed to the retrieval system's ability to provide contextually relevant information, enabling the language model to generate accurate and semantically coherent responses.

These results highlighted the benefits of integrating RAG systems with language models. The retrieval component significantly enhanced factual accuracy and response correctness while maintaining high semantic coherence. Thus, RAG systems improve the reliability of the returned information and can be extended to documents not included in the language model's pretraining dataset [59].

B. RQ2: Can SRLMentor Accurately Recognize Students' Goals and Plans, and How Does Its Precision Compare With Human Assessors?

We conducted Cohen's kappa analysis to evaluate the agreement between the two coders regarding *SRLMentor*'s ability to accurately detect the goals and plans of 25 students across two rounds of goal-setting and planning activities, encompassing 46 records of students' previously set goals and plans.

For goal detection, the two coders concluded that the system correctly identified 37 records, nearly correctly recognized four records, partially correctly identified one record, and correctly detected only a minor portion of one record. For the remaining three records, the goals were inaccurately identified or not detected. Near-perfect agreement was observed between the two coders' judgment, $\kappa = 0.86$, 95% CI [0.67, 1.05], $p < .001$.

Regarding plan detection, the coders agreed that 38 records were correctly identified, two records were nearly correctly recognized, two records were partially correctly detected, one record was partially identified, and three records were not detected. The agreement between the coders was also nearly perfect ($\kappa = 0.75$, 95% CI [0.47, 1.02], $p < .001$). All disagreements were resolved through discussions between the coders.

Cohen's kappa results indicated an overall high level of agreement between the two coders for both goal and plan detection. Descriptive statistics showed that the average goal detection accuracy was 4.54 [standard deviation (SD) = 1.110], whereas the mean plan detection accuracy was also 4.54 (SD = 1.130). The high agreement between the coders and overall accuracy of *SRLMentor* in detecting goals and plans suggested that the system was generally effective. The minor inaccuracies indicated areas where the system could be further refined, such as enhancing NLP algorithms to handle complex or ambiguous student inputs and establishing a feedback system where educators and students can report and review detection errors.

C. RQ3: What Is the Quality of the Overall Feedback Provided by SRLMentor?

Regarding *coherence*, the two coders agreed that the feedback was coherent in the 36 dialogue records. However, the remaining ten records exhibited partially coherent feedback due to the system repeating the same sentence. The agreement between the coders was nearly perfect ($\kappa = 0.87$, 95% CI [0.70, 1.04], $p < .001$). The average coherence score was 2.76 out of 3 (SD = 0.431).

For *relevance*, substantial agreement was observed between the coders ($\kappa = 0.66$, 95% CI [0.03, 1.28], $p < .001$). The mean relevance score was 2.96 out of 3 (SD = 0.206). In addition, perfect agreement was observed between the coders in maintaining a *positive tone* across all 46 dialogue records.

These findings highlight the strengths of the *SRLMentor* system in providing coherent, relevant, and positively toned feedback and identify areas for improvement, such as reducing repetitive responses to enhance coherence. The repetition of certain phrases in some records could be due to a flaw in the dialogue flow logic that causes the system to loop and repeat certain phrases.

D. RQ4: Does Higher Engagement With SRLMentor Lead to Greater Improvements in Students' SRL Skills and Learning Performance?

1) K*-Means Clustering Approach:* Before conducting cluster analysis, we examined the relationships among the log variables to determine whether they could be included together. The Shapiro–Wilk test indicated that only *Turns_GoalPlanning* showed no significant deviation from normality ($p = 0.586$), whereas the remaining log data were nonnormal ($p < 0.05$). Consequently, we used Spearman's rank-order correlation to test for collinearity among variables. As shown in Fig. S9 in Supplementary Material, the results revealed no significant correlations between any pair of log variables (all $p > 0.05$). The correlation coefficients ranged from –0.33 to 0.34, suggesting that each variable captured distinct aspects of students' behavioral engagement supporting their inclusion in the cluster analysis without redundancy.

Before clustering, all log variables were standardized using Z-scores to ensure that they were comparable. We then ran a k-means procedure with $k = 2$ multiple times and obtained consistent results. This process identified two distinct engagement profiles. Fig. 7 displays the cluster centers for each log variable on the y-axis (average Z-scores), and the x-axis represents the two clusters.

1) *Cluster 1 (low engagement):* Cluster centers for all recorded variables were negative, indicating that students in this group ($N = 10$) exhibited relatively low usage of SRLMentor features.
2) *Cluster 2 (high engagement):* Cluster centers for all recorded variables were positive, suggesting that students in this group ($N = 15$) engaged in frequent use of SRLMentor features.

2) Self-Reported SRL Skills: To investigate the association between engagement levels and self-reported SRL, we compared the pre- and posttest OSLQ scores across the two behavioral engagement clusters. Descriptive statistics for these scores are presented in Table II.

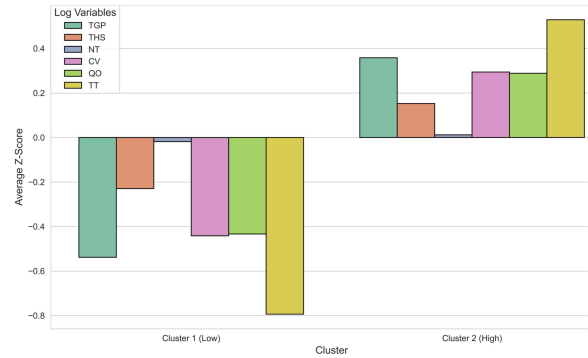

Fig. 7. Two-cluster solution based on students' log variables. Bars represent the cluster center (average Z-score) for specific variables.

TABLE II
DESCRIPTIVE STATISTICS FOR PRE- AND POSTTEST OSLQ IN LOW- AND HIGH-ENGAGEMENT CLUSTERS

Cluster	Phase	N	Min	Max	Mean (SD)
Low	Pre	9	2.08	4.13	3.31 (0.65)
	Post	6	3.00	4.21	3.45 (0.47)
High	Pre	15	1.71	3.71	3.16 (0.46)
	Post	15	2.75	4.96	3.65 (0.53)

In this study, we employed nonparametric tests such as the Mann–Whitney U and the Wilcoxon Signed Rank tests to analyze the data. These methods are particularly well suited for small samples [60], [61], [62], [63]. Their validity with small samples has been further supported by previous studies [63], [64], [65], [66], [67]. By utilizing these methods, we ensured the statistical rigor and appropriateness of our analyses, even with a limited sample size. The Mann–Whitney U test indicated no initial statistically significant differences between the two clusters ($U = 51.50$, $Z = –0.96$, and $p = 0.347$). These findings suggested that the self-reported SRL baselines of the students in the low- and high-engagement clusters were comparable.

We conducted related samples Wilcoxon signed-rank tests on the pre- and posttest scores for each group to assess statistically significant SRL gains within each cluster. For the low-engagement cluster, the median pretest score was 3.50 and the median posttest score was 3.33. The test revealed no significant difference between the pre- and posttest scores ($Z = 0.52$ and $p = 0.600$) in the low engagement group. These results suggested that low-engaged students did not demonstrate significant SRL gains. This may be due to their limited tendency to apply SRL strategies, leading to minimal development of SRL skills.

By contrast, the high-engagement cluster showed a statistically significant increase in OSLQ scores from pretest to posttest. The median pretest score was 3.21, and the median posttest score was 3.50. The Wilcoxon signed-rank test indicated a significant difference between the pre- and posttest scores ($Z = 2.56$ and $p = 0.011$). In addition, the effect size ($r = 0.66$) reflected a large effect, highlighting the practical significance of the observed improvement in SRL among highly engaged students.

TABLE III
DESCRIPTIVE STATISTICS FOR PRE- AND POSTTEST KNOWLEDGE IN LOW- AND HIGH-ENGAGEMENT CLUSTERS

Cluster	Phase	N	Min	Max	Mean (SD)
Low	Pre	10	4	46	21.92 (17.74)
	Post	10	9	43	23.80 (12.40)
High	Pre	15	4	50	25.13 (13.86)
	Post	14	20	76	41.64 (17.51)

3) Learning Performance: We examined knowledge test performance to determine whether higher engagement with *SRLMentor* was associated with improved learning outcomes. Descriptive statistics for pre- and posttest scores across the low- and high-engagement clusters are presented in Table III.

For pretest scores, a Mann–Whitney U test revealed no significant initial difference between the two clusters ($U = 85.50$, $Z = 0.59$, and $p = 0.567$). This indicated that students in both clusters began with comparable prior knowledge.

For the posttest knowledge scores, the Mann–Whitney U Test revealed a statistically significant difference between clusters ($U = 115.50$, $Z = 2.67$, $p = 0.006$, and $g = 1.14$). These findings indicated that students with higher engagement with *SRLMentor* achieved significantly greater mastery of course concepts than their less-engaged peers.

4) Summary: A K-means two-cluster analysis revealed that students who engaged more frequently with *SRLMentor* features, such as goal planning, help seeking, note taking, completing optional quizzes, and overall chatbot interactions, demonstrated improvements in both SRL skills and course knowledge.

From a pedagogical perspective, these findings highlight that GenAI-support tool, such as *SRLMentor*, can effectively foster SRL development and learning. The lack of significant changes in pre- and postactivity OSLQ scores among the low-engagement group may suggest that these students overestimate their SRL abilities [23] and are less inclined to actively apply SRL strategies. Future designs could incorporate gamification elements to motivate students to actively apply SRL strategies. Educators may create incentive structures that reward consistent engagement and improvement in SRL behaviors.

By integrating log data, such as goal-setting checklists, quiz completions, and help-seeking behaviors, with subjective self-report questionnaire data, this study mitigated some of the limitations of relying solely on self-reported measures. Although the k-means-clustering-based analysis provided preliminary evidence that higher behavioral engagement with *SRLMentor* was associated with greater improvements in SRL skills and knowledge outcomes, it cannot substitute for a randomized controlled trial. The lack of a control group weakens the strength of causal conclusions.

E. RQ5: What Are the Students' Perceived Usefulness of the System?

We analyzed interview responses of ten students to examine their perceptions of *SRLMentor*'s usefulness and gather suggestions for future improvements. Two key themes emerged from the interview data, highlighting the usefulness of *SRLMentor*.

1) Clarifying Goals and Plans: Seven students reported that *SRLMentor* helped them clarify their course intentions, reflect deeply on their goals and plans, and review the completion of previous goals.

"The bot really helped me. It gave me a reason to think through things more clearly, what I want to learn in this course, and how I want to learn it." (Student W)

"I really like it personally. I enjoy the feeling of being asked at the end if I have completed everything. This kind of reflection not only provides a sense of closure but also allows me to recognize my progress and achievements." (Student H)

2) Monitoring Learning Process: Seven students stated that *SRLMentor* helped them monitor their learning, manage their time, and review the learning content.

"Yes, it [the bot] does help. It's like having to regularly report my progress to someone. Without this accountability, maybe we could just slack off, couldn't we?" (Student M)

"It [the bot] helps me better plan my time because it asks about my schedule, and even though I feel pressured to respond, I ultimately create a timeline. This helps me better manage my learning progress." (Student C)

Despite the overall positive perceptions of *SRLMentor*, the students reported several limitations that highlighted the need for further improvement.

3) Tediousness of Answering Chatbot's Questions: Several students found it tedious to answer chatbot questions.

"Talking with it can be a bit tedious. It tries to guide you step by step, but you might be in a hurry. It keeps asking you to elaborate on your plans, which can sometimes be a bit frustrating, perhaps because I am a bit lazy." (Student J)

Possible suggestions for addressing this limitation included implementing a user profile system that could adapt question complexity based on user interactions and remember preferences to minimize repetitive questions.

4) Basic Guidance for Advanced Users: Several students reported that the chatbot provided the most basic guidance, which may not be helpful to users seeking advanced help.

"The operational answers that it provides are only effective for the most basic steps. However, when I ask about specific adjustments, like how to fine-tune parameters when adjusting categories, the advice it gives isn't very useful. It can't provide me with very specific operational guidance." (Student L)

This could be addressed by enhancing the chatbot's knowledge base with detailed guides, implementing a tiered response system for complex queries, and integrating human-expert assistance for continuous improvement.

5) Repetition of Replies: Five students found it annoying that the chatbot repeated the same replies.

"There were times when it gives the same response over and over, and I get confused." (Student S)

Possible suggestions to address this included using more advanced NLP algorithms to vary responses based on context and history and integrating a feedback loop to learn from user reactions and improve interactions.

In our current implementation, repeated prompts were intended to reinforce core SRL strategies (e.g., asking students to elaborate their plans). However, as learners' SRL skills improve, the same repetition becomes counterproductive and is experienced as tedious. According to cognitive load theory, redundant or repetitive information can increase extraneous cognitive load [68], potentially distracting learners from core learning tasks. To address this, we will introduce learner control in future refinements of the system to help students manage cognitive load and personalize the interaction. Learners will be offered options to receive detailed chatbot guidance or turn it OFF if they wish. Allowing learners to request chatbot guidance at their own pace can increase learner autonomy and reduce extraneous load.

VI. Conclusion

This article presents *SRLMentor*, an AI system designed to aid students in SRL. We demonstrated that the RAG component enhanced the factual and semantic accuracy of the responses. In addition, *SRLMentor* showed strong alignment with human evaluations of students' goals and plans. The system provided valuable feedback and clear guidance, leading to a substantial improvement in students' SRL, with a large effect size. Practically, educators can leverage this research to enhance students' SRL skills.

While *SRLMentor* has shown promising results, it has several limitations. First, its knowledge base primarily depends on curated local documents, which might limit the breadth and timeliness of information, especially in fast-evolving fields. This reliance could also affect its adaptability across diverse educational settings and cultural contexts. Second, important aspects of SRL, such as emotional regulation and motivation, are not comprehensively addressed. In addition, the relatively small sample size of the current study may limit the generalizability of the findings.

Despite these limitations, the core design principles of SRLMentor were intentionally built for generalizability and contextual adaptation. Although this study involved a small and culturally homogeneous cohort, the system's architecture supports expansion across disciplines, educational levels, and cultural settings. The MASP component encodes the core phases of SRL (forethought, performance, and self-reflection) as digital indicators that can be localized through educator-authored templates containing editable placeholders for learner goals, plans, and monitoring methods. The templates can be customized to reflect disciplinary norms and cultural expressions without modifying the underlying model. The RAG module further enhances adaptability by drawing on educator-curated materials such as textbooks, rubrics, and policy documents, producing responses grounded in local content rather than knowledge fixed within model parameters. Together, these mechanisms provide theoretically consistent yet context-responsive guidance, maintaining SRLMentor's pedagogical coherence across varied learning environments.

Future enhancements to the system include integrating Internet search capabilities and accessing up-to-date academic databases to enhance the system's feedback quality. Advanced NLP techniques, such as sentiment analysis and emotion recognition, will be incorporated to better support the emotional dimensions of SRL. These improvements aim to enhance *SRLMentor's* effectiveness, ultimately boosting users' academic performance and satisfaction.

Future research could benefit from expanding the study to include larger more diverse samples and test the system across different academic disciplines. As previously mentioned, we used nonparametric tests to analyze the data because these methods are well suited for small sample sizes. Although the small sample may constrain the generalizability of the findings, it does not compromise the validity of the results. Instead, our findings provide valuable insights, serving as a critical foundation for future research with larger samples. This approach aligns with the iterative nature of scientific inquiry, where initial small-scale studies pave the way for more extensive investigations. In our present study, we selected coherence, relevance, and positive tone for their practical utility in evaluating feedback quality. Future work should integrate additional frameworks, such as learner-centered feedback principles. This would provide a more nuanced understanding of how feedback supports SRL. In addition, including a control group in future research would allow for a more precise assessment of *SRLMentor's* impact, providing stronger evidence of its overall efficacy. Such studies would help identify potential differences across disciplines and offer valuable insights into the AI system's applicability in various educational contexts.

Acknowledgment

A large language model (e.g., ChatGPT-4o) was used to improve the readability and clarity of this article. After using this tool, the authors edited the content and assumed full responsibility for its accuracy.

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Sikai Wang received the M.Sc. degree in engineering from the University of Hong Kong, Hong Kong, in 2023, where he is currently working toward the Ph.D. degree in educational technology.

His research interests include learning sciences, natural language processing, data mining, and artificial intelligence for education.

Xinyi Luo received the M.Sc. degree in education from the University of Hong Kong, Hong Kong, where she is currently working toward the Ph.D. degree in education technology.

Her research interests include technology-enhanced learning, online learning, and student engagement.

Khe Foon Hew received the M.A. degree in instructional design and technology from Nanyang Technological University, Singapore, in 2002, and the Ph.D. degree in instructional systems design from Indiana University, Bloomington, IN, USA, in 2006.

He is currently a Professor of Education with University of Hong Kong, Hong Kong. His research interests include student engagement and self-regulated learning in online activities.